Digital Design and Computer Architecture I

Module Description

Module code	CS102-F26	**Credits**	7 ECTS
Programme	Foundation programme	**Level**	Semester 1
Semester	Fall 2026	**Prerequisites**	None
Module leader	Dr. Ali Khudiyev	**Email**	ali.khudiyev@wosm.academy
Office hours	Monday–Friday, 10:00–17:00 (Baku time, UTC+4)		

1. Module Overview

This course is for people who are curious about inner working principles of computers. You will learn about computer systems (e.g. basic architecture and organization, Von-Neumann-Computer, machine-instruction cycle, hardware-software interface), instruction set architecture (ISA), fundamental methods of circuit design, structural organization of processor using circuit elements, processor design variants and their implications, some advanced aspects of description and implementation and verification of digital circuits, and other things such as microprocessor architectures and systems, parallel and distributed systems, memory systems, and input/output (I/O) systems.

2. Aims

- TODO(owner): aims of the module.

3. Learning Outcomes

After completing this course, you will be able to:

LO1. Understand computer systems and layered abstract machines.

LO2. Gain an introductory overview of computer architecture and fundamental knowledge of low-level programming, digital circuits, and processor design.

LO3. Apply basic concepts of computer architecture and explain machine instruction cycles based on the underlying hardware at the register-transfer level.

LO4. Classify different types of computer architectures and understand the basic principles of modern computer architecture.

4. Syllabus

Unit	Topic
1	Introduction & Data Representation
2	Assembly Languages
3	Von Neumann Architecture
4	Other ISAs
5	System Architecture
6	Memory Management & Cache
7	Boolean Algebra
8	Adders
9	Multiplication
10	Memory
11	Processor Architecture
12	Finite Automata
13	Multi-cycle Processor
14	Modern Systems & Parallelism
15	The I/O System
16	Physical Layer Implementation
17	Pipelining
18	K-Maps
19	Binary Decision Diagrams
20	SAT
21	Physical Design
22	Review

5. Teaching and Learning Methods

The module is self-paced. All materials are provided on Moodle.

Activity	Hours
Lectures	TODO(owner)
Labs / Tutorials	TODO(owner)
Independent study	TODO(owner)
Total	**TODO(owner)**

6. Assessment

Your understanding and knowledge of the concepts presented will be assessed through quizzes, projects, and challenges. Any permitted support materials will be provided during the examination; no additional aids or resources are allowed.

TODO(owner): weight of each assessment component and the learning outcomes it covers.

7. Reading List

- David A. Patterson, John L. Hennessy. *Computer Organization and Design: The Hardware/Software Interface.*
- Sarah L. Harris, David Harris. 2022. *Digital Design and Computer Architecture: RISC-V Edition.* Elsevier.
- Andrew B. Kahng, Jens Lienig, Igor L. Markov, Jin Hu. 2022. *VLSI Physical Design: From*

Graph Partitioning to Timing Closure. Springer International Publishing.

- RISC-V ISA

8. Materials and Licence

Course materials are published under the Creative Commons Attribution-ShareAlike 4.0 licence unless stated otherwise. See wosm.academy/licence for details.

9. Policies

- **Attendance:** TODO(owner)
- **Late submissions:** TODO(owner)
- **Academic integrity:** All submitted work must be your own. See the Code of Conduct.